

Test Automation

Backend

1st edition

Module 1: API testing

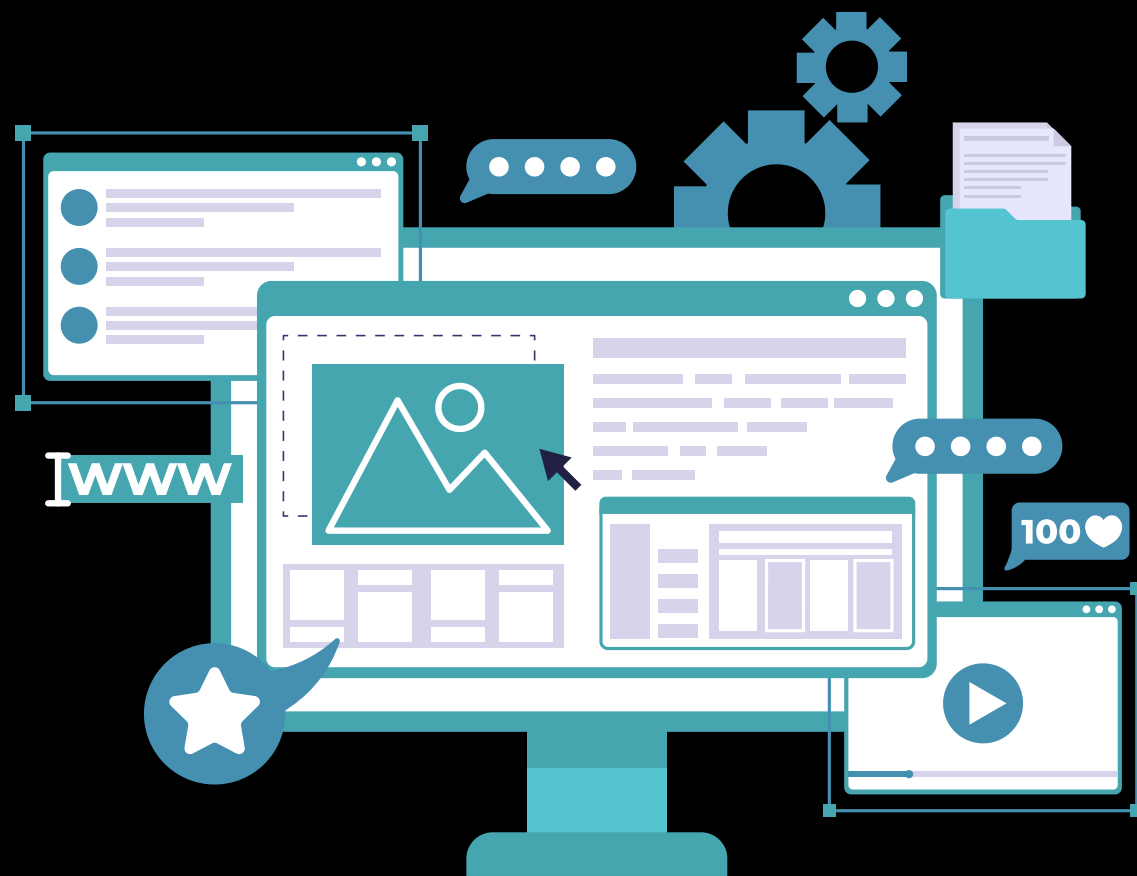
Bruno Machado

APIs Upside Down



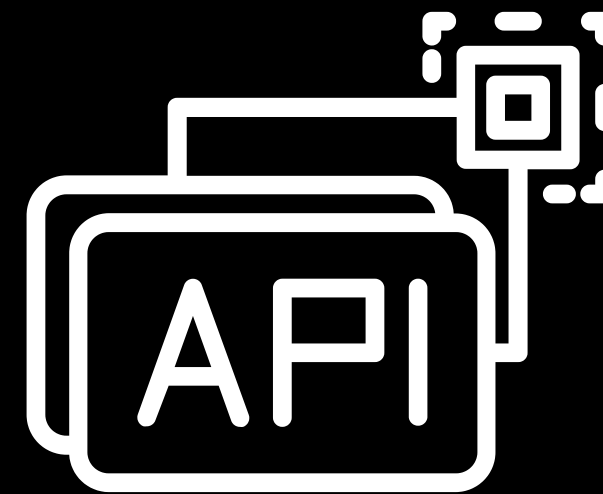
FRONT END

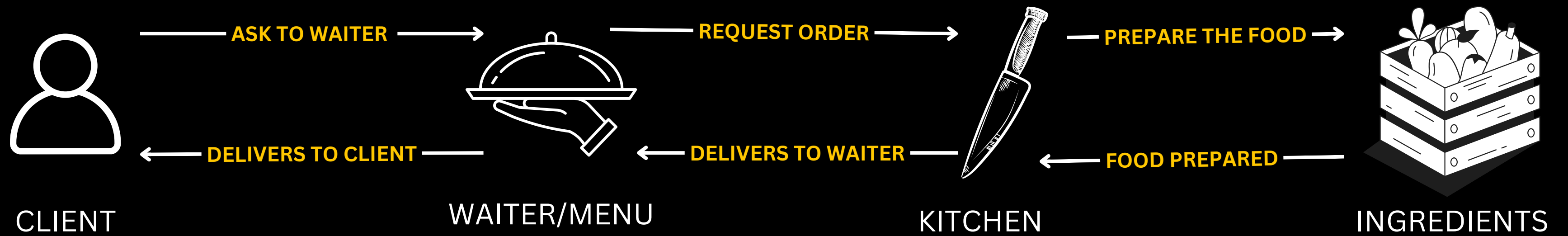
The frontend is the **visible part** of the application or website; that is, everything the user sees and interacts with directly.



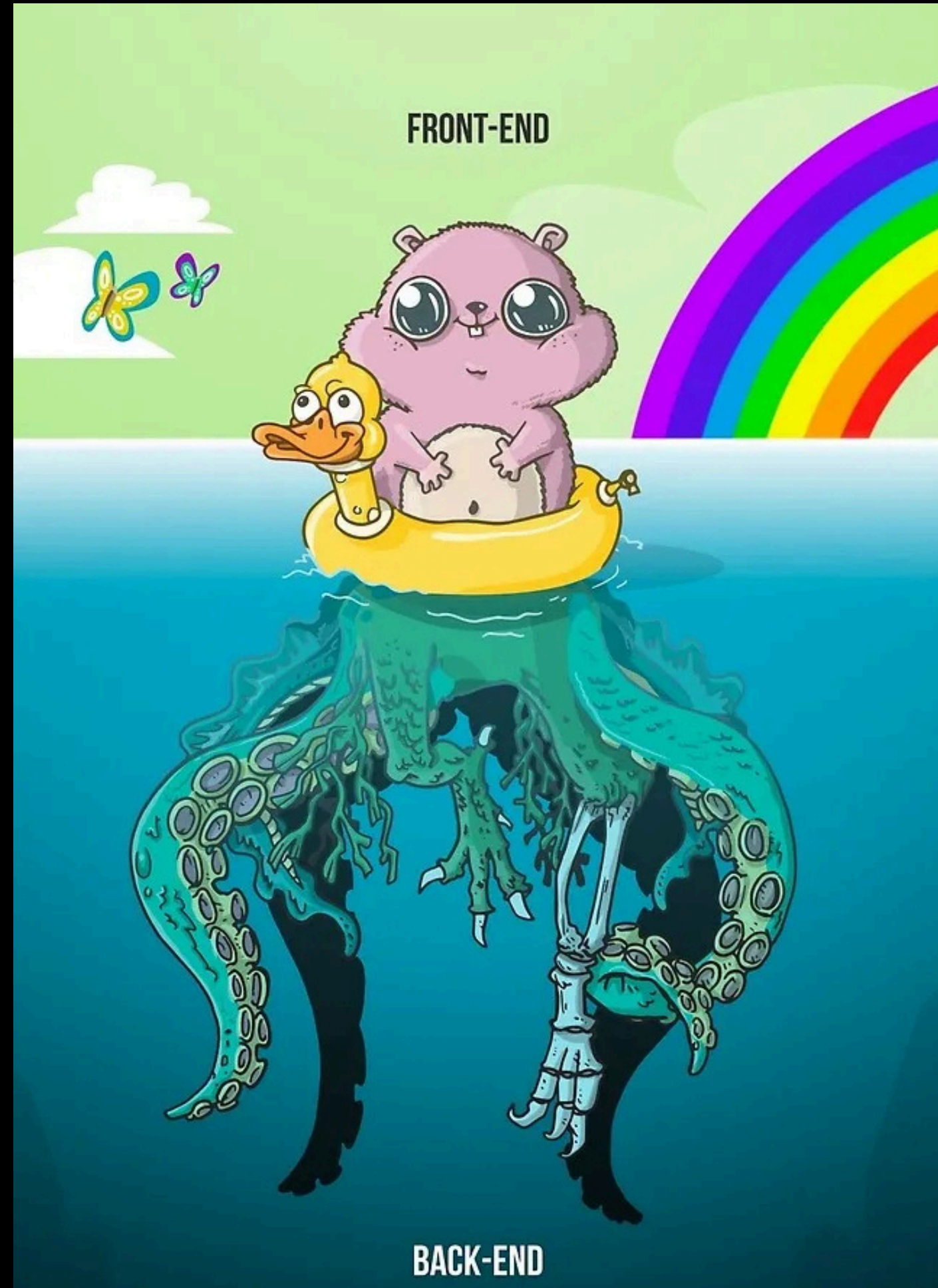
BACK END

The backend is the part of the system that works behind the scenes and manages the **logical operations, data, and interactions** between the server and the database.





FRONT-END



BACK-END

API Types

REST

(Representational State Transfer)

The **most common** type of API, based on web standards, which uses the HTTP protocol to make requests.

REST APIs are known for their simplicity and scalability, being widely used in modern applications.

SOAP

(Simple Object Access Protocol)

An **older protocol** that defines strict rules for communication.

SOAP is heavier and more complex than REST, but it offers greater security and reliability, being used in more conservative legacy systems.

GraphQL

A newer type of API that allows the client to specify **exactly which data it wants to receive**, making responses more efficient.

Unlike REST, which returns a fixed set of data, GraphQL offers **greater flexibility**.

JSON

JSON (JavaScript Object Notation) is a lightweight data interchange format used to store and transmit information between a server and a client.

```
{
  "character": "Eleven",
  "age": 14,
  "isAlive": true,
  "abilities": ["telekinesis", "psychic powers", "mind control"],
  "location": {
    "city": "Hawkins",
    "state": "Indiana",
    "dimension": "Upside Down"
  }
}
```


REST API



- HTTP Methods
- Headers
- Request
- Response
- Status codes

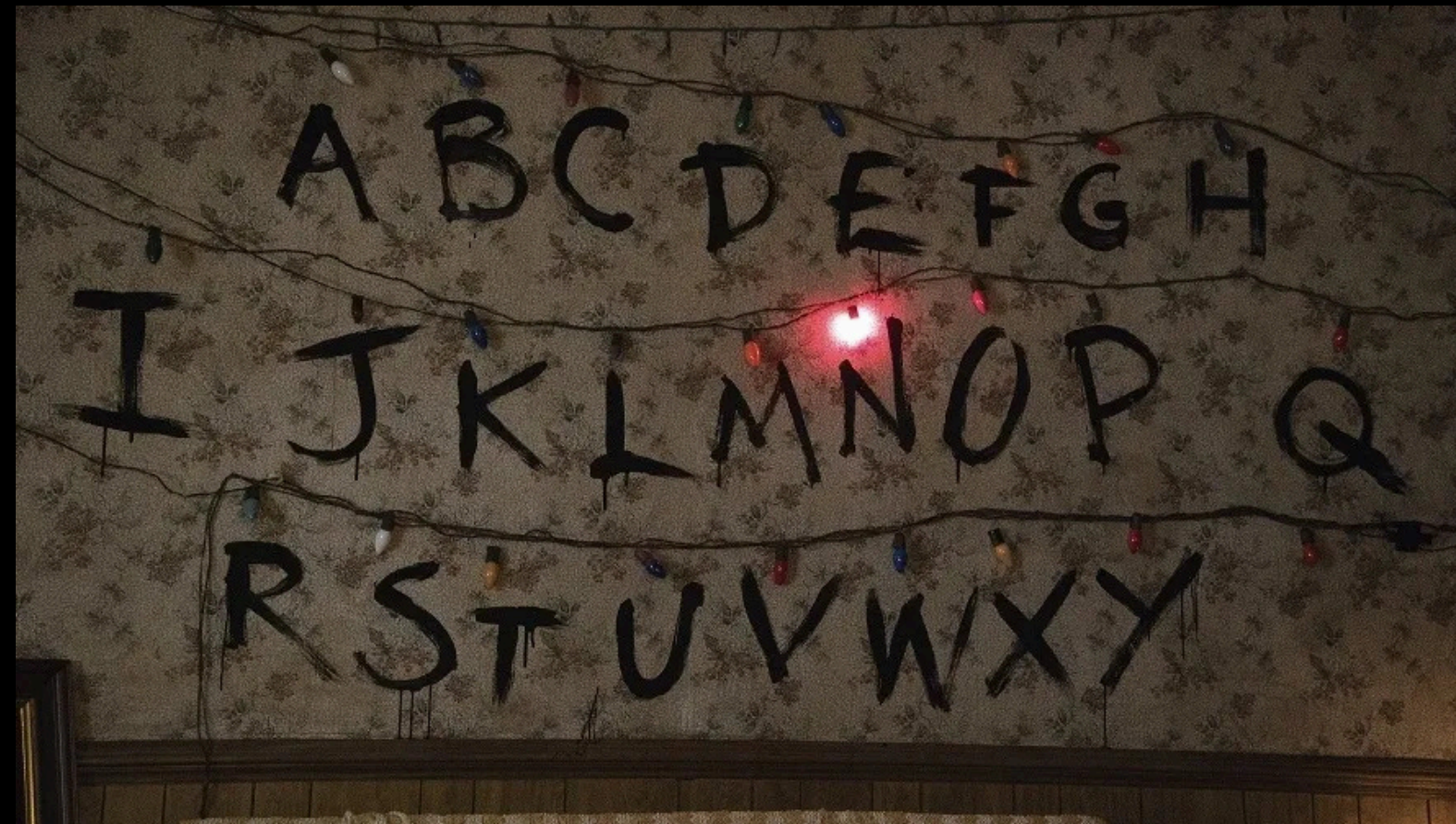
HTTP Methods

HTTP Methods help communicate which **action** you want to perform with the data on the web.

GET	POST
PUT	DELETE
PATCH	HEAD
OPTIONS	TRACE
CONNECT	

GET

GET is used to **retrieve or view data** that is already available, such as viewing a product page, a list of movies, or user details, without modifying or adding anything.



GET

ENDPOINT

GET /api/characters/will-byers

RESPONSE

```
{  
  "name": "Will Byers",  
  "status": "Missing",  
  "location": "Upside Down",  
  "notes": "Trapped in the Upside Down, unable to communicate"  
}
```


STATUS CODE

Status codes are numbers returned by the server to indicate the result of a request. They help communicate whether the operation was successful or if there was a problem.

1xx	INFORMATIVE	Request received and being processed.	100 Continue
2xx	SUCCESS	Request sucessfull	200 OK, 201 Created
3xx	REDIRECT	Request needs more actions to be continued.	301 Moved Permanently
4xx	CLIENT ERROR	Invalid request made by the client.	400 Bad Request, 404 Not Found
5xx	SERVER ERROR	Server failed to process this request.	500 Internal Server Error